VLMs + Adaptive Conformal Inference

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Problem

Given a data-driven/physical model of robots $f_\ell(x, u)$, we can quantify the nonconformity of the model using the score

$$s_{i,\mathrm{mdl}} := \max_{t \in [0,T]} \|\dot{x}_i(t) - f_\ell(x_i(t), u_i(t))\|. \tag{1}$$

And then, sending task specification prompts $\mathscr{L}$ to any VLAs/VLMs, we get target trajectories $(x_d(t), u_d(t)),\ t \in [0, T]$

$$(x_d(t), u_d(t)) = \Phi_{\mathrm{vla}}(\mathscr{L}).$$

We can quantify the validity of the trajectories by a human designed metrics $\mathcal{M}$?

$$s_{i,\mathrm{vla}} := \mathcal{M}(\mathscr{L}_i).$$

Problem

Given a safe set $\mathcal{S}$, let us consider a modified safe desired trajectory $x_d^*(t)$ constructed through the pointwise projection

$$x_d^*(t) \coloneqq \arg\min_{\bar{s} \in \bar{\mathcal{S}}(\epsilon)} \|\bar{s} - x_d(t)\|, \ \ t \in [0, T] \tag{2}$$

$$\bar{\mathcal{S}}(\epsilon) \coloneqq \left\{ \bar{s} \in \mathcal{S} \mid \min_{s_c \notin \mathcal{S}} \|\bar{s} - s_c\| \geq \epsilon \right\}$$

where ϵ is a safety buffer. We then quantify the dynamical inconsistency of the modified desired trajectories $(x_d^*(t), u_d(t))$ by

$$s_{i,\mathrm{dyn}} \coloneqq \sup_{t \in [0,T]} \|\dot{x}_{d,i}^*(t) - f_\ell(x_{d,i}^*(t), u_{d,i}(t))\|. \tag{3}$$

Problem

With conformal inference, we can argue that

$$\begin{aligned} \dot{x} &= f_\ell(x,u) + d_1(x,u) \\ \dot{x}_d^* &= f_\ell(x_d^*,u_d) + d_2(x_d^*,u_d) \end{aligned}$$

with

$$\begin{aligned} \|d_1(x,u)\| &\leq q_{1-\delta_1,\mathrm{mdl}} \\ \|d_2(x_d^*,u_d)\| &\leq q_{1-\delta_2,\mathrm{dyn}} \end{aligned}$$

hold with finite probability, where $q_{1-\delta_1,\mathrm{mdl}}$ is the $(1-\delta_1)$-th empirical quantile for the score (1) and $q_{1-\delta_2,\mathrm{dyn}}$ is the $(1-\delta_2)$-th empirical quantile for the score (3).

Problem

We know that with Lyapunov/contraction + conformal inference, we can guarantee

$$\|x(t) - x_d^*(t)\| \leq e^{-\alpha t}\|x(0) - x_d^*(0)\| + c \cdot \frac{q_{1-\delta_1,\mathrm{mdl}} + q_{1-\delta_2,\mathrm{dyn}}}{\alpha}(1 - e^{-\alpha t}), \ \forall t \in [0, T] \quad (4)$$

for some c with finite probability, where $q_{1-\delta,\mathrm{mdl}}$ is the quantile for the score (1).

We could then either tighten the Lyapunov/contraction condition by the error amount or select the safety buffer ϵ in (2) to be greater than the right-hand side of (4).

Problem

Finally, there will be the sim-to-real gap, which we can quantify using adaptive conformal prediction using the defined scores. The theory should be almost identical to what we did in our CDC paper[1].

[1] I. Kim and H. Tsukamoto. "Closed-loop conformal inference with VLM parameter prediction for nonlinear robust adaptive control". In: *submitted to IEEE Conference on Decision and Control*. Mar. 2026. DOI: TBA.

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Real-World Demonstration

Anyway, our objective is to simulate these guarantees with our robots.

Note that minor details of the theoretical guarantees do not matter; ***we start from experiments, and then interpret the results using our theory***.

Real-World Demonstration

(1) Provide a prompt $\mathscr{L}$ specifying high-level robotic mission objectives and requirements.

- For example, a prompt could be: "*Hey robots, explore the area efficiently and safely, report your findings, and avoid causing harm to humans.*"

(2) Control our robots using the approaches (or simpler versions of the approaches) listed in the previous section.

Note also that ***we shouldn't overcomplicate experimental setups***, as proving our point is just what we need. One thing we can't compromise on is that ***experiments have to involve multiple heterogeneous robots and humans***.

Example Prompt

We may need several iterations of prompt engineering, but here is a prompt that worked reasonably well for ChatGPT.

```
I have three micro UAVs, one UAV with a vision camera, three flappers,
and one hexapod in an environment with 22'x22'x8' motion-captured
space.
Randomly generate cluttered environments with humans and construct
desired trajectories for each of the agents.  When generating random
environments, make sure they are diverse enough so that we can benefit
from the heterogeneity of the robots.
Make sure the trajectories are safe, performant, and most importantly,
agent-aware, meaning you consider each robot's unique capabilities.
For example, hexapods excel in robust terrain exploration on
challenging surfaces, while UAVs specialize in aerial exploration.
Small UAVs and robots can navigate confined spaces, enhancing SLAM
precision through collective intelligence.  Flapping-wing drones are
```

Example Prompt


```
suited for environments with a high risk-to-reward ratio due to their
collision and disturbance resilience.
Multiple flapping drones transporting a hexapod to rugged, higher
elevations can perform detailed multi-level terrain exploration in
high-risk aerial environments.
Just give me numerical results of the environments and trajectories,
not explanations.
```

Example Prompt

Improving the quality of the output prompt is not our point.

Real-World Demonstration

We need the following:

(1) Planning of heterogeneous ground robots in a grid space, with objectives specified solely through natural-language prompts and visual perception (Suman and Ting-Wei?)

(2) Planning and control of homogeneous UAVs and Flapper Drones, with objectives specified solely through natural-language prompts and visual perception, and states represented exclusively by semantic data signals (Inyoung)

(3) Planning and control of heterogeneous ground and aerial robots, with objectives specified solely through natural-language prompts and visual perception, and tasks allocated according to their heterogeneous capabilities (Inyoung)

Real-World Demonstration

(4) Planning and control of heterogeneous ground and aerial robots in the presence of humans, with objectives specified solely through natural-language prompts and visual perception, states represented exclusively by semantic data signals, and tasks allocated according to their heterogeneous capabilities (Inyoung)

Real-World Demonstration

The wording here may be confusing, but just to reiterate, ***what we need is basically multiple heterogeneous robots moving with foundation models with formal guarantees***.

So let's focus on getting things done, even by simplifying problem settings, rather than being too precise in theory.

Real-World Demonstration

Figure: Aerial and ground robots with onboard GPUs and an off-board GPU server. (Image credit: Flapper Drones, bionicbird, ModalAI, Hiwonder, Holybro, DJI, Wheeltec, Robolink, Bitcraze, Dell, and Nvidia. See https://hirotsukamoto.com/robots/.)